ALIVE: Abstract Guidance

Domain, problem, and proposed project description

Based on **ALIVE: Adaptive Life in Virtual Environments - Learning Within Lifetimes, Evolving Across Generations**. The revised proposal focuses on the interaction between learning and inherited adaptation, rather than the earlier reciprocal-coevolution comparison.

This is a planning outline, not a finished abstract or the assignment submission.

Paragraph 1 - Domain

Artificial life and evolutionary computation, specifically how digital agents adapt through lifetime learning and evolution across generations.

Cover these ideas:

- Digital agents operate in simulated environments with limited resources.
- **Lifetime learning** changes an individual agent's behavior through experience.
- **Inherited adaptation** changes the traits passed through generations through inheritance, selection, and mutation.
- Controlled simulations allow researchers to study how these two processes interact as environments change.

Key distinction: learning changes behavior within a lifetime; evolution changes inherited designs across generations.

Paragraph 2 - Problem

How environmental change affects the interaction between learning and inherited adaptation in agents with constrained, modular abilities is the specific issue being investigated.

- Whether combining learning and inherited adaptation improves sustained population performance under slow versus fast environmental change.
- Whether learning preserves diverse inherited abilities or favors a narrower set of flexible designs.
- The difficulty of distinguishing improvements that are inherited from improvements that require individual experience.
- Whether module prerequisites, energy costs, and restricted combinations influence those outcomes.

Research Question & Approach

Connecting the problem to the proposed experiment

Central research question

How does the speed of environmental change affect the interaction between lifetime learning and inherited adaptation in agents with modular abilities?

Avoid claiming that the relationship between learning and evolution is unexplored. The proposal's intended contribution is studying that relationship under **explicit modular constraints and controlled rates of environmental change**.

Paragraph 3 - Project description / approach

Build ALIVE as a controlled simulation that separates and compares the effects of learning and inherited adaptation.

Include:

- A JAX-based simulation of agents competing for finite, regrowing resources.
- An initial design with eight ability modules and four continuous traits.
- Four experimental conditions: neither lifetime learning nor cumulative inherited adaptation; learning only; inherited adaptation only; and learning combined with inherited adaptation.
- Stable environments and slowly or rapidly alternating abundance-scarcity environments.
- Two primary measurements: **sustained population performance** and **inherited module diversity**.
- Tests of archived inherited designs, with learning enabled and disabled, to distinguish inherited improvement from experience-dependent benefits.

If space permits, mention the comparison with module prerequisites removed to test their contribution.

Intended outcome

A reproducible explanation of when learning and inherited adaptation complement one another, create trade-offs, or provide no measurable advantage - **not a promise that the combined condition will win**.

Source: ALIVE_Learning_and_Evolution_Revised_Proposal.docx, revised September 2, 2026. This guide summarizes the proposal; experiments and results are not yet complete.